Philip Baah Afriyie

AI/ML Engineer · Speech, NLP & Applied AI

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Summary

AI/ML Engineer with nearly three years at Ayadata, where I built the Twi speech models offered through AyaSpeech (Beta): Whisper for speech recognition (roughly 7% word error rate), a bidirectional Akan–English NLLB translation model (BLEU 81) and a custom text-to-speech voice. I design, fine-tune, evaluate and serve machine learning and deep learning models, and build LLM applications — agentic RAG and AI agents on LangChain and LangGraph, with Claude on AWS Bedrock — plus the APIs and serving layer around them in Python, FastAPI, Kafka, Redis, PostgreSQL and NVIDIA Triton. I partner with our cloud engineer on cluster provisioning and deployment, and I am comfortable owning a system from a research notebook through to a production-ready handoff.

Experience

Machine Learning Engineer — Ayadata GH

Nov 2023 – Present · Accra, Ghana

- **Speech AI (ASR, TTS) and NLP: fine-tuning, evaluation and serving.** Fine-tuned Whisper (ASR) on real, code-switched farmer speech — English brand names and numerals mixed into Twi — reaching roughly 7% word error rate after building a text-normalization pipeline to clean inconsistent Akan orthography across thousands of transcripts. Fine-tuned a bidirectional Akan–English NLLB-200 model to a BLEU score of 81, tracked in MLflow alongside chrF++ and ROUGE, and trained a custom Akan text-to-speech voice. Packaged all three models for serving on a low-latency NVIDIA Triton layer, built a FastAPI gateway in front of it (auth, rate limiting) and a custom latency/throughput benchmarking dashboard; our cloud engineer provisioned and ran the Triton cluster. The three models are offered through AyaSpeech, Ayadata's Asante Twi speech API (Beta).
- **Conversational voice AI: real-time survey platform (prototype, not deployed to production).** Designed an unattended IVR system that conducts spoken telephone surveys in a local Ghanaian language: speech recognition, machine translation, answer classification, dynamic question routing, and speech synthesis in one pipeline. Built two front ends over a shared survey engine — a LiveKit voice agent for real-time calls with energy-based speech segmentation, and a FastAPI service for batch processing — designed to run behind Gunicorn and nginx with disk-backed session state so any worker could resume any call in progress.
- **LLM agents and retrieval-augmented generation (RAG).** Built agentic RAG systems on LangChain and LangGraph for autonomous context retrieval, writing the Kafka producers/consumers for high-throughput event processing and the Redis-backed conversational state and caching layer; our cloud engineer provisions and operates the underlying clusters.
- **Information extraction and data pipelines (prototype, not deployed to production).** Developed an end-to-end pipeline that ingests raw PDFs and scanned images and converts them into structured PostgreSQL records via OCR, with downstream agents running analytics on the extracted data to generate reports automatically and cut manual administrative work.
- **Agentic performance analytics platform (prototype, not deployed to production).** Built a dashboard that aggregates business and operational metrics into real-time risk scoring and generates LLM-written performance insights for HR and management, with a multi-channel notification service (Slack, email, push) and role-based access control.

- **Computer vision for agriculture.** Engineered object detection models on drone imagery for automated crop counting, reaching 91.8% precision on banana trees with MMDetection and 94% precision on palm trees with Ultralytics YOLOv8, to improve harvest planning.

Selected Projects

Akan–English Speech Pipeline — Fine-Tuning & Serving

Hugging Face Transformers, PyTorch, NVIDIA Triton Inference Server, FastAPI, MLflow, Terraform
The models are offered through AyaSpeech (Beta): app.ayaspeech.ayadata.ai

- **Models and serving:** Whisper (STT), a bidirectional Akan–English NLLB-200 model (MT) and a custom Akan text-to-speech voice, packaged for serving behind an NVIDIA Triton server with a FastAPI gateway in front of it (auth, rate limiting), writing the Terraform/Ansible configs our cloud engineer applies to provision the cluster — validated with a dashboard benchmarking latency and throughput.

Shelby — personal AI assistant

Python, TypeScript, FastAPI, React, MCP, PostgreSQL, Docker, Claude (Anthropic API, AWS Bedrock), Whisper, Pocket TTS
github.com/Philips-DR/shelby

- **My design, built with Claude Code as a pair programmer.** I defined the system architecture, the rules every part must follow and the features myself: six standalone tools, each its own process reached over the Model Context Protocol and sharing no code or credentials; a web app kept separate from the API; containers, a private Tailscale network and PostgreSQL for memory and the audit log; and the rule that nothing is written, sent or deleted without my approval. I reviewed and tested every change.
- **What it does:** a chat assistant that works through those tools (meeting transcription and minutes with Whisper, Markdown to Google Docs, Gmail, checking a tailored CV against the job posting, reading web pages, and PDF-to-audiobook narration in a cloned voice), with voice input and spoken replies, reachable from my phone. An 8-bit speech model and two worker processes made audiobook narration 1.7× faster with no loss in Whisper read-back accuracy, and about 1,700 automated tests run with no network, credentials or model.

Voice AI Survey Platform — real-time conversational IVR

Python, FastAPI, LiveKit, Claude on AWS Bedrock, Gunicorn, nginx
github.com/Philips-DR/voice-ai-survey-platform

- A phone-call-style survey agent that understands free-form spoken answers and branches to the next question with no human operator on the line, via five answer-parsing strategies selected per question type — keyword matching, fuzzy scoring, regex extraction, live LLM entity extraction, and post-call theme coding — on a declarative question graph with conditional branching and a time-based rule that shortens an overrunning call.

Skills

- **Programming & ML frameworks:** Python, PyTorch, Hugging Face Transformers, Scikit-learn, SQL, Bash, TypeScript, JavaScript
- **Speech AI & NLP:** ASR (Whisper), machine translation (NLLB), text-to-speech, text normalization, model fine-tuning, OCR (Tesseract, PaddleOCR)
- **LLMs, RAG & AI agents:** LangChain, LangGraph, agentic RAG, retrieval-augmented generation, Claude on AWS Bedrock, LLM entity extraction
- **Computer vision:** MMDetection, Ultralytics YOLOv8, object detection
- **APIs & web serving:** FastAPI, REST APIs, OAuth 2.0, Gunicorn, nginx
- **MLOps & infrastructure:** NVIDIA Triton (model config/serving), Docker, Kubernetes, MLflow, Terraform, Ansible (IaC authoring), Kafka, Redis (application/producer-consumer layer)

- **Cloud & data:** AWS (EC2, S3, Lambda, Bedrock), PostgreSQL
- **Tools:** Git and GitHub, Power BI, QGIS, VS Code

Education

BSc. Information Technology — BlueCrest University College
Accra, Ghana · CGPA 3.63

Certifications

Natural Language Processing Specialization (DeepLearning.AI, Coursera: Classification and Vector Spaces, Probabilistic Models, Sequence Models, Attention Models) · **AWS Cloud Technical Essentials** (Coursera) · **Conversational AI Expert Author** (DRUID AI) · **Foundations: Data, Data, Everywhere** (Google Data Analytics, Coursera)